

Goodman’s Purity of Pure Project

Verification, Corrections, and Remaining Problems

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This report describes an Isabelle formalization of Jeremy Goodman’s project notes on Purity of Pure in higher-order metaphysics. It presents the verified object-language results and model constructions, explains corrections and qualifications to the arguments in the notes, and identifies the open problems. The formalization builds on the independently maintained Bacon-Dorr development of higher-order logic and Classicism.

Goodman’s consistency question remains open. We have verified substantial object-language arguments and exact-model constructions, including the failure of L2 for the complete closed-logical stock, the rebuilt M5 model, and a corrected M5 collision argument. We have not constructed a model of the full background with the required instance of Purity of Pure, or derived a contradiction from that background. The remaining mathematical obligations are explicitly retained as contributor projects.

The software, source correspondence, and detailed claim ledger are in the [Goodman application folder](#), with the background logic supplied by the [Bacon-Dorr core repository](#) [3, 4]. The application shares the main repository’s visibility and access permissions. The report gives the mathematics rather than duplicating the repository’s implementation guide. Source labels T1–T9 and M1–M7 refer to Goodman’s July 2026 notes [1].

1 The question and the relevant distinctions

Let t be proposition type and put $u = t \rightarrow t$. For each type σ , Pure_σ and Fun_σ have type $\sigma \rightarrow t$. The instance of Purity of Pure used in the central arguments is

$$\text{PP}_u := \text{Pure}_{u \rightarrow t}(\text{Pure}_u).$$

It is one type-indexed instance, not the assertion of PP at every type. Purity of Fun is a different principle and is not assumed in the question.

For pure operators $X, Y : u$, define

$$\text{fun}'(p) := \forall X Y ((\text{Pure}(X) \ \& \ \text{Pure}(Y) \ \& \ Xp = Yp) \rightarrow X = Y).$$

Thus evaluation at p separates pure unary operators. This is not a definition of $\text{Fun}(p)$. Let

G be the pure operators with a pure two-sided inverse, and define

$$X \approx Y := \exists Z \in G (X = Y \circ Z).$$

Composition is on the input side. Identity at each type is the object-language identity, not merely agreement of truth values.

For clarity, let $\mathcal{B}_0$ denote the added logical-purity and application-closure schemas over the represented CEV+ calculus. Logical purity says that every closed term containing only logical vocabulary denotes a pure entity; application closure says that applying a pure function to a pure argument yields something pure. Let $\mathcal{B}_R$ additionally include exactly one fundamental proposition, no fundamentals at other types, and zeroary and unary Recombination. Let $\mathcal{B}_Q$ add the corresponding Exhaustion principles. These distinguish the Recombination-only variant from the full QLN version of the question. In the latter, the target is consistency of $\mathcal{B}_Q + \text{PP}_u$, without Purity of Fun. Results for the weaker variant must be identified separately.

Only zeroary and unary QLN are nonvacuous when there is exactly one fundamental entity and fundamental arguments must be pairwise distinct. This reduction of higher-arity instances is a mathematical observation about the stipulated setting, not an additional cataloged theorem.

The implementation distinguishes ordinary consequence from an axiom extension in which theorem-level rules operate above the added axioms. In particular, a temporary assumption $\text{fun}'(r)$ cannot simply be promoted to a permanent axiom. Nor may a biconditional established under a temporary hypothesis automatically be converted to an identity by the theorem-level Rule of Equivalence. The M5 correction below depends on this distinction.

Proposition-valued Modalized Functionality, for $X, Y : \sigma \rightarrow t$, is derived in the represented CEV development. Its arbitrary-result-type extension is verified in the exact semantics but must not thereby be called redundant over bare CEV. Likewise, model validity and object-language derivability remain different claims.

2 Scope of the formal verification

The development establishes whole-proof forward preservation into the named axiom-extension calculus, with its typing, signature, and closed-axiom conditions explicit. This is not an unqualified two-way identification of every presentation. The exact-model development proves the requisite interpretation and global soundness conditions, including

preservation of the complete closed-logical denotational stock. Truth preservation of sentences alone would not establish that stock correspondence.

All 303 supplied theory files are selected in the verified build. There are 1,708 integration catalog entries in 87 audit catalogs, with nine historical replay entries counted separately. The audits check for oracle dependencies, residual kernel hypotheses, and unresolved unification pairs. These counts are not counts of independent mathematical results, and a clean audit does not erase explicit theorem premises. Source fidelity requires the separate comparison of those premises and definitions with the publications.

The semantic results use Isabelle/HOL with HOL-ZF. Their consistency consequences are relative to that foundational setting. The exact appendix construction is Bacon’s finite-natural-word instance, with its constrained function domains and the chosen singleton individual domain, not a theorem about every monoid or every individual carrier. Its function domains satisfy Bacon’s stated action-compatibility condition. More general core completeness questions are documented in the core’s own source guide.

2.1 The object-language results

The following table summarizes the main numbered claims. The claim ledger supplies their exact endpoints and axiom packages.

Item	Verified content and qualification
T1	With zeroary Exhaustion, pure propositions are truth or falsity. The associated collapse of biconditional operators is verified.
T2a-c	Closure of fun' under pure reversibles, nontriviality of fun' propositions, and attainment of each pure proposition are verified. The displayed fun' antecedents remain explicit.
T2d-f	Possible purity, false-but-possible noncontingency, and the six pairwise-distinct propositions are verified in their stated scopes. Possible purity has a proof not requiring Persistence; the six-distinctness claim retains $\text{fun}'(r)$.
T3	The advertised modal argument reaches possible identity where identity is needed. The repaired heredity claims with zeroary Exhaustion or the specified pure-identity rigidity principle are verified, retaining the stated QSS and purity/persistence conditions. A defect in the modal argument is not a countermodel to the whole CEV+ theory.

Item	Verified content and qualification
T4	For the stated pure $C : t \rightarrow (t \rightarrow t)$, the higher-type failure of fun' at $C(r)$ is verified. The represented result needs logical purity and application closure, not PP or $\text{fun}'(r)$.
T5	The proliferation conclusion is verified as a conditional theorem with the required $\text{fun}'(r)$ antecedent. It is not an existence or consistency proof for that antecedent.
T6	All four contradiction routes are verified with their L2 and classification assumptions retained. Native formulations of the extra principles and the repaired fundamental-witness routes are included.
T7	T7a's absorption result is verified with its L2 assumptions. T7b remains underspecified: the kind-level term, binders, and fixed/shifted relations are not uniquely supplied by the notes.
T8	The five base kinds, kind uniqueness under L2, and the stated 31-element growth bounds are verified. This is not by itself an unbounded iteration proving infinitely many kinds.
T9	The abstract conditional counting theorem and a substantive exact-carrier instantiation, including conditional infinitude, are verified. The arbitrary-model connection remains open; see Section 6.

There are two further completed results concerning the notes' preliminary claims. First, for each type σ and each finite length n , pure parameters $a_1, \dots, a_n$ determine a pure predicate

$$P_n = \lambda x. (x = a_1 \vee \dots \vee x = a_n), \quad P_0 = \lambda x. \perp,$$

with the expected material membership biconditional. The proof is an explicit induction, using the closed logical function $\lambda a. \lambda Q. \lambda x. (x = a \vee Qx)$ and application closure. The empty case is separately verified. Neither PP nor Exhaustion is needed. This proves finite pure comprehension, not comprehension for arbitrary infinite external subsets. Its weaker axiom package inherits relative consistency from the verified stronger no-PP QLN background.

Second, for $B_a = \lambda p. (p \leftrightarrow a)$, the identity $B_a \circ B_a = \text{id}$ is verified with no added axioms. The companion purity and group-membership implications from $\text{Pure}(a)$ are separately packaged over the PP core. That is the package used in those proofs, not a claim that PP is necessary.

The witness $R_+ = \lambda p. (p \& \text{fun}'(p))$ has a native proof of purity, instantiation, and rigid

specification. Consequently TU yields RS over $\mathcal{B}_0 + \text{PP}_u + \text{Exhaustion}_0 + \exists p \text{ fun}'(p) + \text{TU}$. The contradiction between the notes' assertion that TU entails RS and their assertion of incomparability remains a source question. Our theorem does not establish that Goodman intended one assertion only in the weaker setting without Exhaustion.

3 The L2 principles and the open consistency question

The notes have one L-principle in weak and strong forms, not a separate L1. Weak L2 says

$$\text{Pure}(X) \& \text{Pure}(Y) \& \text{fun}'(p) \& \text{fun}'(q) \& Xp = Yq \rightarrow X \approx Y.$$

Strong L2 requires the same $Z \in G$ also to satisfy $q = Zp$. The verified T6 consequences, writing the minimal added schemas explicitly, are

$$\begin{aligned} \mathcal{B}_0 + \text{PP}_u + \exists \text{fun}' + \text{L2} + \text{Inv} &\vdash \perp, \\ \mathcal{B}_0 + \text{PP}_u + \exists \text{fun}' + \text{L2} + \text{WI} &\vdash \perp, \\ \mathcal{B}_0 + \text{PP}_u + \exists \text{fun}' + \text{L2} + \text{TU} &\vdash \perp, \\ \mathcal{B}_0 + \text{PP}_u + \text{strong L2} + \text{RS} &\vdash \perp. \end{aligned}$$

Here $\exists \text{fun}'$ abbreviates $\exists p \text{ fun}'(p)$. These are axiom-extension derivations, not unrestricted uses of a deduction theorem.

The repaired route from fundamentality derives $\exists \text{fun}'$ using Recombination, zeroary Exhaustion, and unique fundamentality. Recombination supplies the unboxed identity information, but the available QSS argument needs the further modal step supplied by zeroary Exhaustion. We do not infer nonderivability of QSS from Recombination alone merely from that gap. The T6 derivations that assume $\exists \text{fun}'$ directly require no QLN axiom.

For a model sound for the relevant axiom extension and globally validating the repaired background with PP, T6 excludes simultaneous global validity of L2 and TU. This is not automatically the claim that one of those formulas is false at the distinguished root. Nor does failure of one make the remaining axioms consistent.

3.1 L2 fails for Bacon's complete closed-logical stock

In the prefix presentation of the finite-natural-word frame, put

$$Z(P) = \{w : (\exists n w^\frown \langle n \rangle \in P) \& (\exists n w^\frown \langle n \rangle \notin P)\}.$$

Word reversal relates this presentation to Bacon’s division action. Isabelle identifies Z with the denotation of a closed logical term in the exact carriers. It proves that Z is surjective, is not injective, and is right-cancellative for composition in the complete closed-logical stock. The definition tests variation among all immediate successors of a world, not just two distinguished children.

For example, taking $P_S = \{w \smallfrown \langle 0 \rangle : w \in S\}$ gives $Z(P_S) = S$. Also $Z(P) = Z(\sim P)$, so Z is not reversible. Composition closure and right-cancellation ensure that if p separates the pure operators, so does Zp . A verified generic construction supplies such a p . But

$$\text{id}(Zp) = Z(p).$$

L2 would make $\text{id} \approx Z$. Since the witnessing input transformation is reversible, this would make Z reversible, a contradiction. The formalization connects this calculation to actual root falsity of the quantified L2 formula in the specified no-PP interpretation. Strong L2 fails there as well.

This answers Goodman’s first model-calibration question for the complete fixed logical stock. It does not show that L2 fails in every enlargement of Pure, or that PP cannot imply L2. Enlarging Pure changes both L2 and fun' . Historical bounded Vampire searches for L2 and sufficient intermediate claims returned no proof in the reported runs. Those failures are not underderivability results, were not rerun for this report, and add no Isabelle certificates to the present verification.

4 The exact model claims and their repairs

M1. At proposition type, the classifier of the complete logical stock is the noncontingency operator. Bottom-type PP is verified. At the next relevant type, the classifier exists in the exact domain and PP is equivalent to its membership in the next pure stock. Its unconditional nonmembership has not been proved. This part of M1 remains open.

Bacon’s footnote-59 diagonal is separately verified, both on the exact carriers and through an arbitrary-carrier compositional interpretation with the explicit CEV+ soundness conditions. The obstruction retains Purity of Fun, QSS, and the fundamental witness with the stipulated uniqueness. A bare H model does not automatically provide that soundness interface. Since Purity of Fun is absent from Goodman’s question, this obstruction is not a negative answer to it.

M2. For each set T of propositions, the classifier $G_T(P) = \{i : i \cdot P \in T\}$ is an invariant unary value, and the construction gives a bijection with the invariant unary values. Cantor's theorem and explicit collisions show why interpreting all invariants as pure is incompatible with QSS at a fundamental proposition. This rejects that interpretation of purity, not logical purity itself.

M3, M4, and M6. Freeness relative to the complete logical stock, both extreme views of a fun' proposition, and the product-meagerness result are instantiated on the exact carriers. M4 has an actual suitable branch-lift witness for each exact-stock fun' input. M6 has actual separation and nonindependence results, including a strict-inclusion pair whose relation is preserved by substitutions. The required premises concerning the complete logical stock are instantiated in these exact-model results.

The condition that every view of one fixed r be fun' is proved impossible and is not assumed in the M4/M6 proofs. It must not be confused with the correctly guarded modal QSS principle. The branch in M4 must be chosen suitably; the arbitrary missing target in the countability part of M6 is not asserted to be fun' .

M5's rebuilt model. The rebuilt interpretation uses Bacon's exact carriers. Its expanded pure stock includes denotations of all closed terms over the additional operator, including abstraction and higher-order quantification, not merely a finite list of applications. The rebuilt fundamental proposition supports the explicitly verified no-PP QLN background and QSS. The displayed exotic operator is pure, self-inverse, and neither uniformly truth-preserving nor uniformly truth-flipping. TU and Inv have root failures; WI nonderivability is established by its theorem-level relation to TU, not an unproved root implication.

Two source qualifications matter. For the displayed no-echo argument, the claim that the only prefixes are the empty word and the word itself uses a word of length one. Further, there are only countably many world-indexed displayed pairs, so countability alone cannot select one outside a countable orbit. A uniform diagonal pair construction supplies the needed pre-rebuild obstruction. The later rebuilt interpretation changes the fundamental proposition rather than leaving an old witness fixed while enlarging Pure. No PP model is obtained.

4.1 The corrected M5 collision

The source also proposes that $F(p) = (p \leftrightarrow \text{NC}(p))$, where $\text{NC}(p) = \Box p \vee \Box \sim p$, has equal values at $\text{NC}(r)$ and $\top$ whenever $\text{fun}'(r)$. That local assertion is false in the verified exact model. Adding $\text{fun}'(r)$ as a permanent axiom does not justify this local claim: the resulting

stock is already inconsistent. Nontriviality gives $r \neq \top$ as a theorem and hence its necessitation, whereas attainment gives $\diamond(r = \top)$. These conflict without PP. A local assumption or the existential sentence $\exists \text{fun}'$ is a different matter.

The repaired input is $q = \text{NC}(\Box r)$. The following is an actual object-language derivation, not merely a semantic collision calculation.

Proposition 1. *Over $\mathcal{B}_0$, CEV+ proves*

$$\text{fun}'(r) \rightarrow (q \neq \top \ \& \ Fq = F\top), \quad \text{fun}'(r) \rightarrow \sim \text{Reversible}(F).$$

Proof. Write $q = \Box\Box r \vee \Box\sim\Box r$. S4 proves $q \rightarrow \Box q$. It also proves $\sim\Box\sim q$: axiom 4 gives $\sim q \rightarrow \sim\Box r$, so normality gives $\Box\sim q \rightarrow \Box\sim\Box r$, hence q . Axiom T simultaneously gives $\Box\sim q \rightarrow \sim q$. Thus $q \leftrightarrow \text{NC}(q)$ is a theorem. Both Fq and $F\top$ are therefore theorems, and theorem-level Equivalence gives $Fq = F\top$ with no added axioms.

Now assume $\text{fun}'(r)$ temporarily. Nontriviality gives $\sim\Box r$; attainment at pure truth gives $\diamond\Box r$. Hence both disjuncts of q are false. By T, $\sim q$ implies $\sim\Box q$, that is, $q \neq \top$. Combine this with the previously established output identity and discharge the temporary assumption. Reversibility implies injectivity, yielding the second conclusion. $\square$

The independent audit checks that this proof does not depend on collision theorems using a fixed fun' axiom or on the resulting collapse. The separate semantic collision at the proposition true everywhere except the root remains valid, but is not substituted for this proof. No general collision method can refute reversibility merely because a candidate fails TU: the rebuilt M5 model has a nonuniform involution with no collision. A PP-specific argument would require further premises.

M7. The complete-stock diagonal is verified on the exact carriers for every typed input R : a proposition exists that is neither pure nor the value of a pure unary operator at R . Reachability by all invariant operators is equivalent to injectivity of $i \mapsto i \cdot R$. For every exact-stock fun' input that orbit map is noninjective, with an explicit unreachable singleton; this includes the verified generic fundamental seed. Identifying a particular intended Theorem-10.1 gluing with such a seed is still a separate contributor problem. No claim about every arbitrary gluing is made.

5 Bacon's appendix metatheory

Proposition 8 and Theorem 10.1 here refer to the appendix of *Logical Combinatorialism* [2], not the introductory book. The exact recursion's all-type closure, action laws, and surjec-

tivity are verified for the represented construction. Native global soundness and the exact complete-stock correspondence are also verified. The supplied QLN interpretation is the specified unique-fundamental-proposition construction; it must not be silently identified with every gluing of constant assignments.

The name-parametric Theorem 10.1 provides, for each countable family $(A_n)_{n \in \mathbb{N}}$ of suitably typed interpretations over an arbitrary type of constant names, one glued interpretation C such that

$$\langle n \rangle \cdot \llbracket M \rrbracket_C = \llbracket M \rrbracket_{A_n}$$

for every closed named term M of the t -generated fragment. The same C works for all such M . The proof codes each term's finitely many constant names into strings; it does not inject the entire signature into strings or infer that the signature is countable. On string names the polymorphic evaluator agrees with the string evaluator. The family of branches remains countable. Open-term transport and types involving e are not included in this extension.

The enumeration and frame-representation proofs are selected, audited, and transferred to named sentences. For a fixed string signature and a closed sentence M in the fragment, the one complete interpretation C satisfies

$$M \text{ is frame-satisfiable} \leftrightarrow C, \langle \rangle \models \Diamond M,$$

$$M \text{ is frame-valid} \leftrightarrow C, \langle \rangle \models \Box M.$$

Frame satisfiability means root truth under some typed interpretation of the constants on the fixed frame; validity means root truth under every such interpretation. The modal forms retain the rich-variable and typed-assignment conditions. This is not syntactic H completeness or an effective decision procedure. The new arbitrary-name gluing theorem does not automatically generalize the countable sentence-enumeration theorem.

6 Goodman's suggested attacks

1. Calibrate L2. The fixed complete closed-logical stock has been calibrated negatively, as explained in Section 3. Whether PP forces L2 in the intended full theory remains open.

2. Cap or characterize G . The classification principles and their T6 uses are verified, as are the M5 counterexamples without PP. The granularity analysis supplies a precise sufficient condition: a surjective proposition operator that preserves sameness of truth value is truth-preserving or truth-flipping, assuming there are both true and false propositions. Reversibility alone does not give this condition.

With unary QLN, logical purity and application closure, the locally guarded agreement test for pure Z and fundamental r gives

$$\text{NC}(Zr \leftrightarrow r) \leftrightarrow (\forall p (Zp \leftrightarrow p)) \vee (\forall p \sim(Zp \leftrightarrow p)).$$

This is an equivalence to the corresponding truth-uniformity condition, not a weaker principle already derived from PP. Unary Exhaustion is part of the reduction. The TU-to-RS result above also retains its stated Exhaustion and existence assumptions.

3. Exclude finitely many kinds under PC. Let K be the kinds and P the pure unary values. The checked exact-carrier interpretation, with its global purity axioms, root L2 and $\exists\text{fun}'$, and full external unary PC, gives

$$|\mathcal{P}(K)| \leq |P| \leq |K \times G|.$$

The additional verified separation argument excludes finite K : if $|K| = n$, classifiers selecting subsets of sizes $0, \dots, n$ occupy different kinds, since the induced reversible action preserves those cardinalities. Thus K is infinite and $2^{|K|} \leq |G|$.

This is real progress beyond the notes' finite-or-exponentially-large dichotomy, but under explicit semantic hypotheses. Full external PC ranges over every external subset of the relevant pure values; it is not silently supplied by an internal comprehension formula. The transfer to arbitrary suitable Henkin models, including the necessary quotient and selector constructions, remains open. No bound $|G| \leq |K|$ has been derived from Goodman's central theory. Such a bound would conflict with the displayed inequality, but the cardinal argument does not supply it.

4. Prove consistency. No model of the full target has been verified. The exact no-PP models show that failures of L2 or TU are possible in specified interpretations; they do not discharge PP's self-classification requirement. The verified finite-support theorem says that inconsistency of the axiom extension would already be witnessed by a finite set of added axioms. It is a proof-theoretic finiteness result, not a construction of models for every finite set or a semantic compactness proof. Exploratory models and unfinished search programs are not included among this report's results.

Goodman's PC also must not be conflated with Bacon's stronger identity premise in the Melianism discussion. Coextension with a pure property does not by itself identify that property with Pure_σ . The relevant infinite-disjunction argument is in Bacon's main text with notes 57–58; note 60 points to the appendix. We retain the classifier-extension result with-

out claiming an independently verified general complete-Boolean-algebra or unconditional classifier-nonpurity theorem.

7 What remains for contributors

The verified development includes the native TU-RS transfer, finite PC at every type, biconditional-operator algebra, named frame representation and necessity, and arbitrary-name closed-term Theorem 10.1. These results do not finish the entire formalization. The following four projects remain open.

1. Resolve unconditional higher-type classifier nonmembership in M1. The existing equivalence with PP does not decide either side.
2. Supply the arbitrary-model T9 construction, retaining the distinction between full external PC and internally represented subsets.
3. Specify the intended glued fundamental proposition and prove the freeness or fun' property needed for its M3/M7 application.
4. Obtain precise formulations of T7b and the multiple-fundamental M4 proposal, and clarify the conflicting TU/RS source passages.

The repository gives bounded starting points for these projects. Further algebra, open-term transport, and broader model constructions are separate extensions, not accomplishments implied by the completed tasks. In particular, the verified scope does not encompass every determinate source claim, and not every theory file has received an exhaustive fidelity audit.

8 Logical distinctions and verification safeguards

The corrected results address the main dangers identified at the end of Goodman's notes. Input negation and output negation are distinct; local truth is not identity; a theorem of a model is not automatically a theorem of the object calculus; substitutions form a monoid, not a group; and fun' expresses separation, not surjectivity of an orbit. The M5 countability error and the global-versus-local collision error are concrete instances, not merely hypothetical warnings.

The present contribution is therefore a verified collection of arguments, repairs, and precise limitations. It gives Goodman exact evidence about which proposed routes work

and where further mathematics is needed. It does not favor either answer to the consistency question by itself.

References

- [1] Jeremy Goodman. *Is Purity of Pure consistent with Bacon's Logical Combinatorialism?* Project notes, July 2026.
- [2] Andrew Bacon. Logical Combinatorialism. *The Philosophical Review*, 2020. Appendix numbering follows the source copy identified in the repository.
- [3] Andrew Bacon. *A Philosophical Introduction to Higher-Order Logics*. Routledge, 2024.
- [4] Andrew Bacon and Cian Dorr. *Classicism*. Draft dated 1 July 2022; the core formalization's source numbering refers to this draft.